

MWESIGYE EMMMY

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SUMMARY

Software and systems-focused IT graduate student with hands-on experience in building scalable applications, cloud-native systems, and data-driven solutions. Skilled in Python, backend development, and working with containerized environments, with practical exposure to machine learning, anomaly detection, and data processing workflows. Experienced in analyzing system data, designing reliable services, and leveraging AI-driven approaches to extract insights. Strong interest in MLOps, automation, and developing robust data pipelines for real-world applications.

EDUCATION

Master of Science in Information Technology

August 2025 - May 2027

Carnegie Mellon University, Africa

- Major: Software engineering
- Minor: Networking Technologies

Bachelor's Degree

April 2021 - June 2025

University of Rwanda, Kigali, Rwanda.

Information Security

PROFESSIONAL WORK EXPERIENCE

Internship

Feb 2021 - July 2021

Cisco Networking Academy online

- Completed intensive networking and IT infrastructure training focused on routing, switching, and enterprise networking concepts.
- Earned CCNA Routing and Switching and Cisco IT Essentials certifications.
- Built foundational skills in network troubleshooting, device configuration, and systems support.

Software Developer

Aug 2023 - June 2024

Rwanda National Police

- Collaborated with the software development team to design, maintain, and improve internal systems and applications.
- Assisted in developing and supporting backend services and system integrations for organizational workflows.
- Participated in deployment, configuration, and maintenance of applications on on-premise infrastructure and internal servers.
- Contributed to system updates, configurations, backups, and documentation to support operational continuity.
- Provided technical support to end users by diagnosing and resolving software, system, and application-related issues.

PROFESSIONAL SKILLS

- **Programming & Web development Skills:** Java, JavaScript, Python, C/C++, Data Structures & Algorithms, React.js, Express.js, Node JS, RESTful API design.
- **Frontend & UI:** React ecosystem, Responsive UI development, Familiarity with Tailwind CSS.
- **Database and Data:** MySQL, MongoDB, PostgreSQL, Basic data modeling & querying.
- **Cloud & DevOps:** Kubernetes, Docker (containerized development), OpenStack, Hadoop
- **Security:** Malware Analysis (RISA), Cybersecurity Fundamentals, Network Security
- **Networking:** Enterprise network design, router/switch configuration
- **Tools:** Git, Linux, CI/CD fundamentals, Linux environments, Prometheus, Grafana, Loki
- **Technical Skills:** MS Excel, MS Word, MS PowerPoint
- **Languages:** Fluent in Kinyarwanda and English, and a little French
- **Interpersonal:** Strong communication skills, cultural competency, and adaptability in science

PROJECTS

Mobile Money Fraud Detection System**Dec 2021 - Jan 2022**

- Developed a mobile money fraud detection Android system with a group of developers that captures fraudsters in real time.

Enterprise Cloud Infrastructure**Aug 2025 - Dec 2025**

- Built an Infrastructure-as-a-Service cloud environment using Kubernetes, Docker, OpenStack, and Hadoop. Configured clusters, deployed workloads, and implemented multi-node virtualization and cloud networking.

Advanced Networking Labs**Aug 2025 - Dec 2025**

- Designed and implemented scalable enterprise network topologies with routing protocols, VLANs, OSPF, NAT, and security configurations.

Malware Analysis Training (RISA)**June 2024 - June 2024**

- Completed hands-on malware analysis labs, including static and dynamic analysis, behavioral analysis, and defensive techniques.

AIOps-Observability with Boutique microservices application**March 2026 - April 2026**

- Completed hands-on AIOps observability labs, using google boutique containerized microservices application using tools; Prometheus (collected metrics), Grafana (built dashboards) and Loki (log aggregation).

Anomaly Detection and Machine Learning in Intrusion Detection Systems**March 2026 - April 2026**

- Designed and implemented a machine learning pipeline for email spam detection using both probabilistic and neural approaches.
- Built a Naive Bayes classifier (MAP and ML) with Laplace smoothing and log-likelihood computations to handle numerical stability.
- Developed a single-neuron neural network (logistic regression) and trained it using gradient descent on extracted features.
- Evaluated model performance using accuracy, precision, recall, and F1-score, achieving over 95% accuracy across models.

Intrusion Detection System (IDS) – Agentic AI-Assisted Security Analysis**March 2026 - April 2026**

- Designed and implemented an intrusion detection system in a multi-tier virtualized network environment using Snort, leveraged an agentic AI tool (OpenClaw) to assist in traffic analysis, rule generation, and system hardening workflows.
- Performed network traffic analysis and reconnaissance to identify services, vulnerabilities, and attack patterns.
- Used packet analysis tools (e.g., Wireshark) to inspect network traffic and validate detection rules

CERTIFICATIONS

- ICDL Profile Certificate (2023)
- Certificate of CCNA Routing and Switching (2021)
- Certificate of Cisco IT Essentials (2021)
- Coursera usable security Certificate (2020)
- Introduction to ArcGIS Pro, ArcGIS Enterprise Fundamentals – ESRI Rwanda (2025)
- Malware Analysis Training - RISA (2024)